

Credit Risk Portfolio Performance Optimisation Using SQL and Power BI

DMAIC Framework (Define – Measure – Analyze – Improve – Control)

DEFINE

Business Problem

This project analysed an unsecured lending portfolio to understand why financial performance remained weak despite relatively stable borrower credit behaviour.

Initial portfolio indicators revealed a disconnect between credit risk performance and financial outcomes. While lending demand remained relatively stable and default performance showed year-over-year improvement, the portfolio continued to generate negative net revenue and negative net present value (NPV), suggesting deeper structural inefficiencies within the lending model.

The analysis was designed to answer the following business questions:

- Why is the lending portfolio underperforming financially despite relatively controlled default behaviour?
- Which borrower segments are driving weak portfolio economics?
- Is underperformance primarily caused by credit defaults, provisioning pressure, pricing inefficiency, or exposure concentration?
- What strategic portfolio actions would improve profitability while maintaining responsible lending practices?

The objective was to deliver an end-to-end credit analytics workflow spanning data engineering, data quality governance, dimensional modelling, KPI engineering, executive dashboard reporting, and strategic portfolio diagnostics.

MEASURE

Data Ingestion and Staging

The raw unsecured lending dataset was first imported into MySQL to establish a controlled analytical environment for transformation and governance.

A staging layer was created to preserve source integrity while enabling structured validation and downstream modelling.

This staging process included:

- datatype standardisation
- trimming and cleaning text attributes
- numeric casting for financial fields
- preparation of application, origination, and maturity dates
- preservation of source-level traceability

This ensured the dataset entered the analytical pipeline in a structured and auditable format.

Data Quality Assessment

Before analysis, a comprehensive SQL-based data quality audit was performed using a consolidated validation framework built with **UNION ALL**.

The objective was to evaluate data reliability against key credit risk business rules using a single validation audit structure.

Validation checks covered:

Completeness Checks

- null application IDs
- null customer IDs
- missing financial metrics
- incomplete date records

Uniqueness Checks

- duplicate application detection

Financial Validation

- debt-to-income ratio outside valid range
- invalid credit utilisation values
- affordability score inconsistencies
- fraud risk score inconsistencies

Date Integrity Checks

- origination before application
- maturity before origination

Credit Risk Formula Validation

Expected loss validation:

[
Expected\ Loss = PD \times LGD \times EAD
]

Profitability Validation

Net revenue validation:

[
Net\ Revenue = Interest\ Income - Funding\ Cost - Provision\ Cost
]

Data Remediation

Validation identified several inconsistencies that required correction before the dataset could be considered analytically reliable.

Issues identified included:

- invalid ratio metrics
- financial logic mismatches
- inconsistent lifecycle dates
- inaccurate expected loss calculations
- incorrect profitability measures

To preserve ETL lineage, remediation was performed in a dedicated transformation layer (**clean_loan_data**) rather than altering staging data directly.

Corrective actions included:

- constraining ratio-based measures to valid business ranges
- correcting chronological inconsistencies across loan lifecycle dates
- recalculating maturity dates based on corrected origination dates and agreed loan term
- recalculating expected loss using standard credit risk methodology
- recalculating net revenue using defined profitability logic

Following remediation, the full validation framework was rerun.

Result:

Zero outstanding validation issues.

This confirmed the dataset as a trusted analytical source.

Dimensional Data Modelling

Once validated, the flat operational dataset was remodelled into a dimensional star schema to improve analytical performance, reduce redundancy, and support scalable reporting in Power BI.

Fact Table

fact_loan_performance

Containing measurable portfolio metrics such as:

- approved loan values
 - exposure at default
 - expected credit loss
 - borrower income metrics
 - debt measures
 - delinquency performance
 - profitability indicators
 - NPV
 - risk-adjusted return
-

Dimension Tables

Business attributes were separated into dedicated lookup dimensions:

- dim_employment
- dim_region
- dim_channel
- dim_loan_reason
- dim_approval
- dim_risk
- dim_delinquency
- dim_loan_status
- dim_date

Surrogate keys were introduced to replace repeated text attributes and enforce relational integrity.

Relationships were configured using one-to-many cardinality.

This improved:

- report performance
- scalability
- maintainability
- analytical consistency

KPI Measurement Layer

A structured DAX analytical layer was built in Power BI.

Measures were organised by the business domain.

Portfolio KPIs

- Total Applications
 - Approved Loans
 - Approval Rate
 - Default Rate
 - Portfolio Exposure
 - Expected Credit Loss
 - Net Revenue
 - Portfolio NPV
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Profitability KPIs

- Interest Income
 - Funding Cost
 - Provision Cost
 - Net Margin
 - Cost of Funds Ratio
 - Average Net Revenue per Approved Loan
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Credit Risk KPIs

- Average PD
 - Average LGD
 - Provision Burden %
 - Average Days Past Due
 - Missed Payments
-

Underwriting KPIs

- Average Approved Amount
 - Average Credit Score
 - Approval Mix by Risk Segment
-

Time Intelligence KPIs

Year-over-year trend measures were created for executive monitoring, including:

- YoY application change
 - YoY approval movement
 - YoY default movement
 - YoY exposure movement
 - YoY expected loss movement
 - YoY profitability movement
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ANALYZE

Portfolio Performance Overview

The descriptive dashboard provided a portfolio-wide view of lending activity, risk exposure, and profitability.

Key portfolio metrics

- Total Applications: **6,314**
 - Approved Loans: **4,279**
 - Approval Rate: **67.8%**
 - Default Rate: **9.7%**
 - Portfolio Exposure: **£50.29M**
 - Expected Credit Loss: **£9.95M**
 - Net Revenue: **-£351K**
 - Portfolio NPV: **-£339K**
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Key Descriptive Findings

Lending demand remained stable

Application volumes showed consistent monthly lending demand, with only marginal year-over-year decline.

Interpretation:

Demand-side weakness was not the primary issue.

Approval appetite remained commercially active

Approval rate remained strong at 67.8%.

Interpretation:

The business continued deploying capital actively.

The profitability issue was not caused by reduced lending activity.

Default performance improved

Default rate improved by 3.9% year-over-year.

Interpretation:

Borrower repayment performance improved.

This disproved the assumption that deteriorating defaults were driving losses.

Expected loss remained elevated

Despite lower defaults:

- portfolio exposure increased by 1%
- expected credit loss increased by 1%

Interpretation:

Improved realised repayment performance was offset by increased capital exposure.

Profitability remained structurally negative

Net revenue and NPV both remained negative.

Monthly net revenue remained below break-even in most periods.

Interpretation:

The lending model was structurally loss-making.

Product profitability weakness was systemic

All loan purposes generated negative net revenue.

Worst-performing purposes:

- Travel
- Medical
- Debt Consolidation

Interpretation:

The issue was portfolio-wide rather than isolated to a single lending product.

Exposure concentration in medium-tenor lending

Over 60% of portfolio exposure sat within 24–36 month loan tenors.

Interpretation:

Exposure concentration may be increasing cumulative credit uncertainty and funding burden.

Diagnostic Root Cause Analysis

The second dashboard identified the structural drivers of weak performance.

Profitability pressure from credit provisioning

Portfolio economics:

- Interest Income: **£11.1M**
- Provision Cost: **£10.0M**
- Funding Cost: **£1.5M**
- Provision Burden: **89.6%**
- Cost of Funds Ratio: **13.6%**
- Net Margin: **-3.2%**

Interpretation:

For every £1 generated:

- 90p was consumed by provisioning
- 14p by funding costs

Total costs exceeded income.

Root cause:

Structural profitability pressure.

High-risk borrowers were not the problem

Average net revenue per approved loan:

- High Risk: **+£1,355**
- Subprime: **+£344**
- Prime: **-£232**
- Near Prime: **-£245**

Interpretation:

High-risk lending generated the strongest unit economics.

Root cause:

Risk-adjusted pricing in high-risk segments appeared effective.

Capital allocation was inefficient

Exposure mix:

- Near Prime: **39.9%**
- Prime: **36.5%**
- Subprime: **21.6%**
- High Risk: **2.0%**

Interpretation:

76%+ of portfolio exposure sat in economically weak segments.

Root cause:

Poor portfolio allocation.

Provision burden concentrated in dominant segments

Provision burden by segment:

- Near Prime: **95.8%**
- Prime: **92.5%**
- Subprime: **83.0%**
- High Risk: **71.9%**

Interpretation:

The most heavily funded segments also consumed the highest share of revenue through provisioning.

Root cause:

Capital concentration + weak economics + elevated credit burden.

IMPROVE

Strategic Recommendations

1. Tighten Near Prime underwriting

Reason:

- highest exposure
- negative unit economics
- highest provision burden

Actions:

- stricter affordability thresholds
 - tighter score cut-offs
 - reduced approval appetite
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2. Reprice Prime borrowers

Reason:

Prime borrowers generated negative economic returns.

Actions:

- APR recalibration
- risk-adjusted pricing review
- segmentation refinement

3. Selective Subprime expansion

Reason:

Positive unit profitability with manageable provision burden.

Actions:

- controlled growth
- performance monitoring

4. Cautious High Risk expansion

Reason:

Strong economics but limited exposure.

Actions:

- carefully monitored expansion
- performance validation

5. Product pricing review

Reason:

Systemic negative profitability across loan purposes.

Actions:

- reassess pricing adequacy
 - align pricing with embedded credit risk
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CONTROL

Monitoring Framework

To support ongoing governance, executive dashboards were developed in Power BI.

Dashboard 1 — Lending Portfolio Performance

Monitors:

- lending demand
- approval performance
- default trends
- exposure growth
- expected credit loss
- profitability
- NPV
- product performance

Purpose:

Portfolio health monitoring.

Dashboard 2 — Portfolio Risk Diagnostics

Monitors:

- profitability drivers
- provision burden
- segment economics
- exposure concentration
- strategic segment recommendations

Purpose:

Management decision support.

Ongoing Monitoring KPIs

Recommended governance metrics:

- Net Margin

- Provision Burden %
 - Default Rate
 - Expected Credit Loss
 - Portfolio Exposure
 - Average Net Revenue per Loan
 - Segment Profitability
 - Approval Stability
 - Risk Mix Concentration
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Final Business Outcome

This analysis demonstrated that the portfolio's weak financial performance was not primarily caused by worsening borrower default behaviour.

The primary drivers were:

- structural provisioning pressure
- inefficient capital allocation
- weak economics in dominant borrower segments
- pricing inadequacy relative to embedded credit risk

Final conclusion:

Portfolio profitability can be improved through tighter underwriting in economically weak segments, repricing of underperforming cohorts, and reallocation of capital toward stronger risk-adjusted opportunities.